

Rishikesh Kumar

Full-Stack Software Engineer

Phone: +91 9341421415 | Email: rishi7258prince@gmail.com

linkedin.com/in/rishikesh-kumar-profile | github.com/Rishikesh-kumar-7258 | rishikesh-kumar-7258.github.io

Education

IIT Bombay <i>M.Tech in Industrial Engineering and Operations Research (IEOR)</i>	Jul 2024 – Jul 2026
Guru Ghasidas University <i>B.Tech in Computer Science and Engineering (CSE)</i>	Jul 2020 – Jul 2024

Experience

Full-Stack Developer Intern <i>Jumps Automation</i> Remote	May 2026 – Present
- Developed CyberAegis, a gamified cybersecurity training platform with 7 microservices and 41 interactive simulators - Engineered 128+ REST API endpoints serving a React 18 SPA with event-driven architecture using RabbitMQ and Redis - Deployed 9 containerized services on Kubernetes; wrote 261+ integration tests validated for 5,000 concurrent users	
Full-Stack Developer Intern <i>Vajra Labs, IIT Bombay</i> Powai, Maharashtra	Aug 2025 – Dec 2025
- Built a malware analysis platform processing 200+ daily submissions with static, dynamic, and memory analysis reports - Engineered sandboxed multi-VM execution with an asynchronous Celery pipeline for high-throughput concurrent analysis - Developed using Python, Django/DRF, and Next.js/TypeScript with JWT auth, TOTP 2FA, CAPTCHA, and rate limiting - Led a 3-member team using Agile sprints, delivering across 2 concurrent projects including the company website frontend	
Full-Stack Developer Intern <i>Presear Software Pvt. Ltd.</i> Bilaspur, Chhattisgarh	May 2023 – Jul 2023
- Developed a full-stack web application using React.js and Django REST Framework with component-based architecture - Shipped production-ready features directly from client requirements as 2-member team lead with iterative release cycles	

Technical Skills

- Languages (Proficient):** C, Python, C++, JavaScript, TypeScript, SQL **(Familiar):** Go, Rust, Java, Bash, Lua
- Frameworks & Libraries:** Django, DRF, FastAPI, React.js, Next.js, Node.js, Tailwind CSS, HuggingFace Transformers, PyTorch
- Systems & Tools:** PostgreSQL, Redis, Docker, Kubernetes, Git, GitHub Actions, nginx, RabbitMQ, Celery, WebSockets, Linux

Projects

The Third Eye: Vintel Platform <i>Polyglot Microservice Cybersecurity News System</i>	Jan 2026 – May 2026
- Architected a polyglot microservice system using Go, TypeScript, Rust, and React with persistent memory and A2A protocol - Designed a multi-agent LLM pipeline with two-round editorial deliberation and Borda-count cross-story ranking algorithm	
VersatAI Lab Website <i>Full-Stack Content Management Platform</i>	Dec 2025 – Present
- Developed a full-stack platform for managing research activities, student profiles, projects, and events with RBAC - Built secure REST APIs with Django/DRF and an SSR/SSG frontend with Next.js; deployed with Docker Compose - Implemented JWT-based authentication, TOTP-based 2FA, equation-based CAPTCHA, and role-based access control	
HeadBall2 Game Player <i>Real-Time Mobile Game Automation with Computer Vision and RL</i>	Feb 2025 – Mar 2025
- Built a real-time computer vision pipeline processing live mobile game video at 30 FPS using OpenCV and Python - Engineered a WebSocket client-server architecture achieving 50ms end-to-end latency for reinforcement learning game control	

Research

Parameter Specific Literature Retrieval <i>Master's Thesis</i> Advisor: Prof. Balamurugan Palaniappan	Jul 2025 – Nov 2025
- Built a biomedical literature retrieval pipeline over 87K+ PubMed abstracts to extract parameter-specific clinical evidence - Achieved entity extraction F1-score of 90% by fine-tuning a PubMedBERT-based NER model on annotated dataset - Deployed a distributed ML pipeline with Ray, PyTorch, and HuggingFace Transformers on a Next.js/FastAPI stack - Built a provenance-tracked Neo4j knowledge graph and QA engine reaching an 88% answer rate with every claim cited	

Leadership

Teaching Assistant <i>IE 201, AI & ML Lab, IIT Bombay</i>	Jan 2026 – Apr 2026
Designed lab experiments, set and graded exams for 40 B.Tech students, and conducted doubt sessions on AI/ML concepts	
Teaching Assistant <i>IE 643, Deep Learning – Theory and Practice</i>	Jul 2025 – Nov 2025
Led weekly programming sessions for 80+ students; managed grading, project planning, and regular doubt sessions	
Website Coordinator <i>IEOR Day, IIT Bombay</i>	Jan 2025 – Mar 2025
Coordinated the annual IEOR Day with a team of 30+ members; developed and maintained the event's official website	

Achievements

- Published** research paper on **Plant Disease Classification** using Deep Learning in **International Journal of Computing**
- Achieved **99.30 percentile (2024)** and **99.13 percentile (2023)** in **GATE CSE** among 200,000+ candidates
- Secured **World Rank 32** in **CodeChef Starters 42** competitive programming contest (Data Structures and Algorithms)
- Secured **Rank 1** in **HackForCG Hackathon** by developing an **ML-based computer vision** solution using **OpenCV** and **Haar Cascade**